

Object Oriented Programming using Python

Interview Questions

1. What is Programming Paradigm and how it is useful?
2. What is difference between procedure oriented programming and object oriented programming
3. What is Object Oriented Programming (OOPs)?
4. Explain about advantages of oops?
5. Explain about object and class with real world example and programs
6. Explains differences between class and object?
7. What is the variable and how many types of variables are there in oops?
8. What is self variable and advantages of self variable in python?
9. What is local variable and reference variables
10. Difference between local and reference variables?
11. What is instance variable and how it is differ from local variables
12. What is class level or static variables in python and how is differ from instance variables
13. Difference between instance variables and class level variables or static variables
14. What is constructor in python
15. Explain advantages of constructor in python
16. Explain about `__init__()` method in python and how does it work?
17. Difference between constructor and method in python
18. Define method and explain different types of methods in a class?
19. What is non static method or instance (non-static) method with example program?
20. What is static method or class method with example program?
21. What is name decorator in python and what is advantage of using it?
22. Explain difference between method and constructor in python oops?
23. What is Data Encapsulation and give different real world examples of data encapsulation in python?
24. What is data hiding in python and how can we achieve it?
25. What is access modifier and explain different types of access modifiers?
26. Explain about public, protected and private access specifier in python?
27. Differences between private and public access specifier in python?
28. Explain differences between static and non static methods in python
29. Explain about wrapper function in decorator function with example program?
30. What is name mangling in python and what is advantages of name mangling?
31. Explain about `dir()` function in python?
32. What is decorator and explain advantages of decorators?

33. Explain different types of available decorators in python
 - i) @classmethod
 - ii) @staticmethod
 - iii) @abstractmethod
 - iv) @dispatch
34. What are the magic methods or dunder methods in python?
35. What is Data Abstraction in python and explain with real world example?
36. How do we achieve the data abstraction in python?
37. What is abstract method?
38. What is abstract class and explain about advantages of abstract classes?
39. What is drawback of abstract class in python?
40. What is difference between abstract method and concrete methods in python?
41. How do we create abstract class in python?
42. Difference between class and abstract class in python?
43. Explain about advantages of abstract classes in python?
44. What is inheritance in python? Advantages of inheritance in python?
45. Explain about parent/super/base class and child.sub/derived class in python?
46. Explain difference types of inheritances in python?
47. Explain about single level, multilevel, hierarchical, multiple and hybrid inheritances with example programs?
48. Explain about MRO in python and what is an advantage of it?
49. Explain about fundamental rules of MRO (Two Important rule)
50. Does python support multiple inheritance concept if yes explain how it is possible?
51. What is polymorphism and explain it with real world example?
52. What is operator overloading in python and explain it with real world examples?
53. What is method overloading in python and explain it with real world examples??
54. What is method overriding in python and explain it with real world examples?
55. What is multipledispatch module in python and why we need to install it?
56. What is PIP command and how does it work and why?
57. Differences between method overloading and method overriding in python?
58. Explain use of super() function in python and explain how does it work?
59. Differences between self and super() function in python?
60. What is error in python and explain different types of errors in python?
61. What is an exception in python?
62. Explain about differences between error and exception in python

63. List some important types of errors with example programs?
64. List some important types of exception with example program?
65. What is Exception Handling in python?
66. Explain about try, except, except-else, finally and raise keywords in python with example programs?
67. Explain about finally clause and explain about advantages of finally block with real world example?
68. What is user defined exception and how we can create user defined exceptions in python?
69. What is important of raise keyword in exception handling concept?
70. Explain about Exception Propagation in python?